

EZHILVENTHAN KULANDAIVEL

DATA SCIENTIST | MACHINE LEARNING

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SUMMARY

Data Science graduate student at the University of Alabama at Birmingham with hands-on experience in machine learning, artificial intelligence, predictive analytics, anomaly detection, computer vision, and data visualization. Skilled in Python, SQL, statistical analysis, data preparation, feature-oriented workflows, model development, and performance evaluation.

TECHNICAL SKILLS

Programming: Python (Pandas, NumPy, Matplotlib), SQL

Machine Learning: PCA, SVD, K-Means, anomaly detection, model evaluation

Computer Vision: YOLOv8, OpenCV

Data & Visualization: Power BI, Excel, Data Cleaning, Data Visualization, MySQL

ACADEMIC PROJECTS

Credit Card Anomaly Detection | Feb 2025 - Apr 2025

- Developed an anomaly detection framework for potentially fraudulent credit card transactions.
- Applied PCA and SVD for dimensionality reduction and pattern discovery in high-dimensional financial data.
- Implemented K-Means clustering to separate normal transaction behavior from anomalies and evaluated unsupervised approaches for risk analysis.

Traffic Monitoring Drone System | Feb 2026 - Apr 2026

- Built a real-time traffic analytics platform using YOLOv8 and OpenCV to detect, track, and count vehicles from drone footage.
- Designed a MySQL data pipeline for traffic and violation data and integrated backend services with FastAPI.
- Developed a JavaFX dashboard with visual reports for traffic patterns and system performance.

E-Commerce Sales Analytics Dashboard | Jun 2026

- Developed an interactive Excel dashboard to analyze sales performance, customer behavior, product trends, and delivery metrics.
- Created KPI reporting, PivotTable analyses, and visualizations to support business decision-making.
- Used data cleaning, transformation, and dashboard design techniques to present actionable reports.

Data Analysis Chatbot | Dec 2023 - Apr 2024

- Built a Python data-analysis application with Streamlit to help users view and understand data through an interactive interface.
- Performed data cleaning and manipulation with Pandas, emphasizing simple and accessible analysis.

EDUCATION

University of Alabama at Birmingham | Master's in Data Science (Current), GPA: 3.2 | Jan 2025 - Dec 2026

Dr. M.G.R Educational And Research Institute | Bachelor of Computer Application in Artificial Intelligence & Data Science, GPA: 7.9/10 | Sep 2021 - Sep 2024

CERTIFICATIONS - COGNITIVE CLASS.ai by IBM

- Introduction to Python | Jun 2023 - Jul 2023
- Data Visualization | Jun 2023 - Jul 2023
- Cloud Application | Jun 2023 - Jul 2023